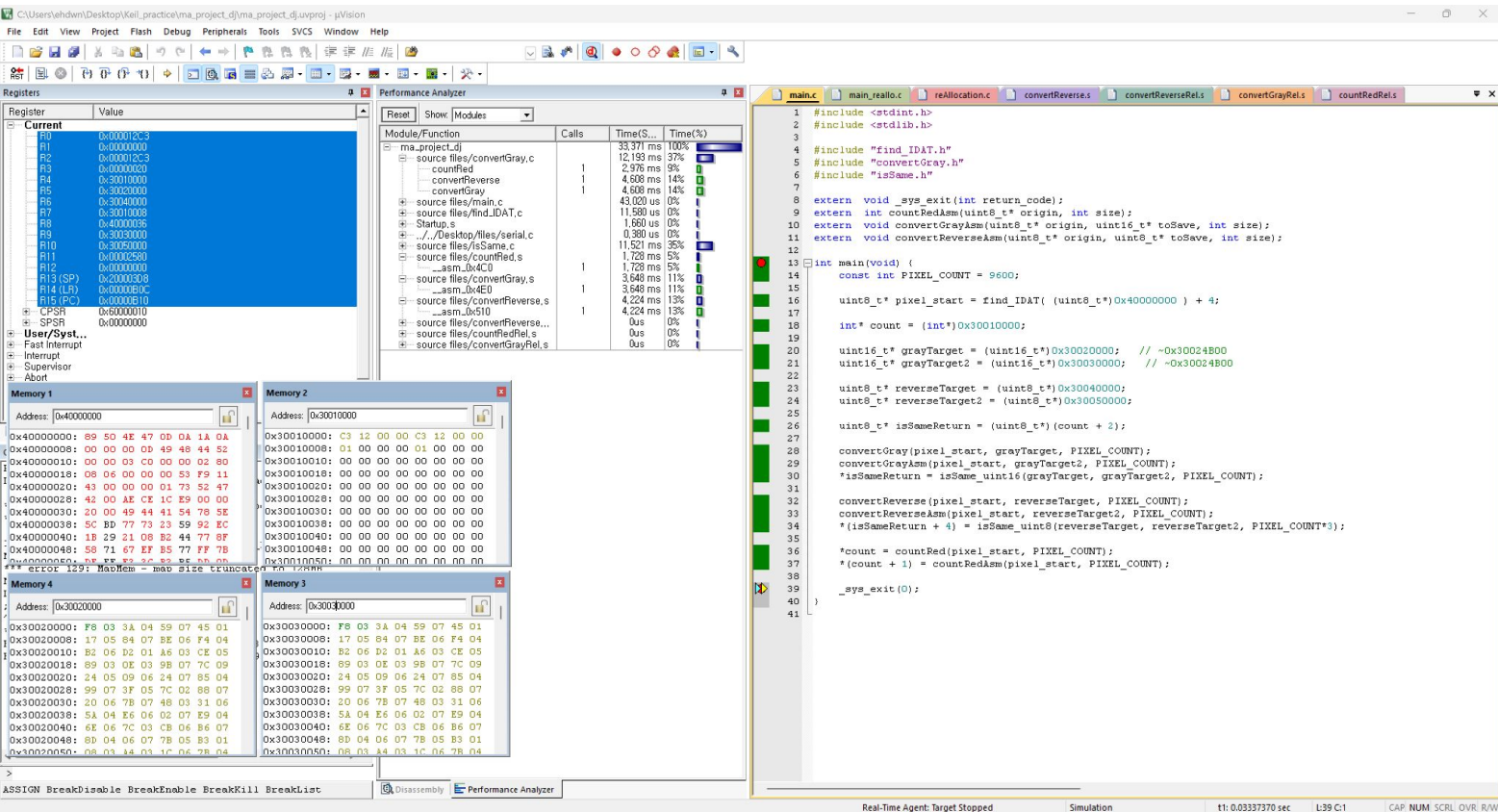


마이크로프로세서 응용 수행 결과

김동준

C언어 & ASM 구현 결과 상호 검증(cnt, gray)



C언어 & ASM 구현 결과 상호 검증 (reverse)

C:\Users\ehdwml\Desktop\Keil_practice\ma_project_dj\ma_project_dj\proj - uVision

File Edit View Project Flash Debug Peripherals Tools SVCS Window Help

Registers

Register	Value
Current	
R0	0x00012C3
R1	0x0000000
R2	0x00012C3
R3	0x0000000
R4	0x3001000
R5	0x3003000
R6	0x3004000
R7	0x3001008
R8	0x4000036
R9	0x3003000
R10	0x3003000
R11	0x0000280
R12	0x0000000
R13 (SP)	0x2000036
R14 (LR)	0x000000C
R15 (PC)	0x0000010
CPSR	0x0000010
+ CPSR	0x0000000
+ User/Sys...	
+ Fast Interrupt	
+ Interrupt	
+ Supervisor	
+ Abort	

Performance Analyzer

Module/Function	Calls	Time(S...)	Time(%)
ma_project_dj		33.371 ms	100%
source files/convertGray.c		12.193 ms	37%
countRed	1	2.976 ms	9%
convertReverse	1	4.608 ms	14%
convertGray	1	4.608 ms	14%
source files/main.c		43.020 ms	0%
source files/find_IDAT.c		11.660 ms	0%
Startup.s		0.380 us	0%
./././Desktop/files/serial.c		11.321 ms	35%
source files/isSame.c		1.728 ms	5%
asm_bx4C0	1	1.728 ms	5%
source files/convertGray.s	1	3.646 ms	11%
asm_bx4E0	1	3.646 ms	11%
source files/convertReverse.s	1	4.224 ms	13%
asm_bx510	1	4.224 ms	13%
source files/convertReverse...		0us	0%
source files/countRed.s		0us	0%
source files/convertGrayRel.s		0us	0%

Memory 1

Address: 0x4000000

0x40000000: 89 50 4E 47 0D 0A 1A 0A
0x40000008: 00 00 00 00 49 48 44 52
0x40000010: 00 00 00 00 00 00 02 80
0x40000018: 08 06 00 00 00 53 F9 11
0x40000020: 43 00 00 00 01 73 52 47
0x40000028: 42 00 AE CE 1C E9 00 00
0x40000030: 20 00 49 44 41 54 78 5E
0x40000038: 5C BD 77 73 23 59 92 EC
0x40000040: 18 29 21 0E 82 44 77 8F
0x40000048: 58 71 67 EF 85 77 FF 78
0x40000050: 8F EF E2 30 83 8E 8B 00
error 129: Memory - map size truncated on 129th

Memory 2

Address: 0x3001000

0x30010000: C3 12 00 00 C3 12 00 00
0x30010008: 01 00 00 00 01 00 00 00
0x30010010: 00 00 00 00 00 00 00 00
0x30010018: 00 00 00 00 00 00 00 00
0x30010020: 00 00 00 00 00 00 00 00
0x30010028: 00 00 00 00 00 00 00 00
0x30010030: 00 00 00 00 00 00 00 00
0x30010038: 00 00 00 00 00 00 00 00
0x30010040: 00 00 00 00 00 00 00 00
0x30010048: 00 00 00 00 00 00 00 00
0x30010050: 00 00 00 00 00 00 00 00

Memory 3

Address: 0x3005000

0x30050000: 87 A1 A3 88 BC DC 6D 13
0x30050008: E4 DE F7 4D 88 70 A7 98
0x30050010: 10 4A 00 84 20 0C C3 4C
0x30050018: 22 62 92 7D FD BF 98 C0
0x30050020: 08 7F 4C E3 56 D5 4B CD
0x30050028: B0 61 3D 3C 3C 06 0F 0E
0x30050030: 4A 82 E8 CC 3A 2D 2E 49
0x30050038: 92 14 CA 79 76 22 2F 23
0x30050040: B0 2E B2 CB A2 32 45 3A
0x30050048: 93 4E 49 0F 4A 92 8B CE
0x30050050: 39 03 2F 3A 32 0C 3B 51

Memory 4

Address: 0x3004000

0x30040000: 87 A1 A3 88 BC DC 6D 13
0x30040008: E4 DE F7 4D 88 70 A7 98
0x30040010: 10 4A 00 84 20 0C C3 4C
0x30040018: 22 62 92 7D FD BF 98 C0
0x30040020: 08 7F 4C E3 56 D5 4B CD
0x30040028: B0 61 3D 3C 3C 06 0F 0E
0x30040030: 4A 82 E8 CC 3A 2D 2E 49
0x30040038: 92 14 CA 79 76 22 2F 23
0x30040040: B0 2E B2 CB A2 32 45 3A
0x30040048: 93 4E 49 0F 4A 92 8B CE
0x30040050: 39 03 2F 3A 32 0C 3B 51

main.c

```
#include <stdint.h>
#include <stdlib.h>

#include "find_IDAT.h"
#include "convertGray.h"
#include "isSame.h"

extern void _sys_exit(int return_code);
extern int countRedAsm(uint8_t* origin, int size);
extern void convertGrayAsm(uint8_t* origin, uint16_t* toSave, int size);
extern void convertReverseAsm(uint8_t* origin, uint8_t* toSave, int size);

int main(void) {
    const int PIXEL_COUNT = 9600;

    uint8_t* pixel_start = find_IDAT( (uint8_t*)0x40000000 ) + 4;

    int* count = (int*)0x30010000;

    uint16_t* grayTarget = (uint16_t*)0x30020000; // ~0x30024B00
    uint16_t* grayTarget2 = (uint16_t*)0x30030000; // ~0x30024B00

    uint8_t* reverseTarget = (uint8_t*)0x30040000;
    uint8_t* reverseTarget2 = (uint8_t*)0x30050000;

    uint8_t* isSameReturn = (uint8_t*) (count + 2);

    convertGray(pixel_start, grayTarget, PIXEL_COUNT);
    convertGrayAsm(pixel_start, grayTarget2, PIXEL_COUNT);
    *isSameReturn = isSame_uint16(grayTarget, grayTarget2, PIXEL_COUNT);

    convertReverse(pixel_start, reverseTarget, PIXEL_COUNT);
    convertReverseAsm(pixel_start, reverseTarget2, PIXEL_COUNT);
    *(isSameReturn + 4) = isSame_uint8(reverseTarget, reverseTarget2, PIXEL_COUNT*3);

    *count = countRed(pixel_start, PIXEL_COUNT);
    *(count + 1) = countRedAsm(pixel_start, PIXEL_COUNT);

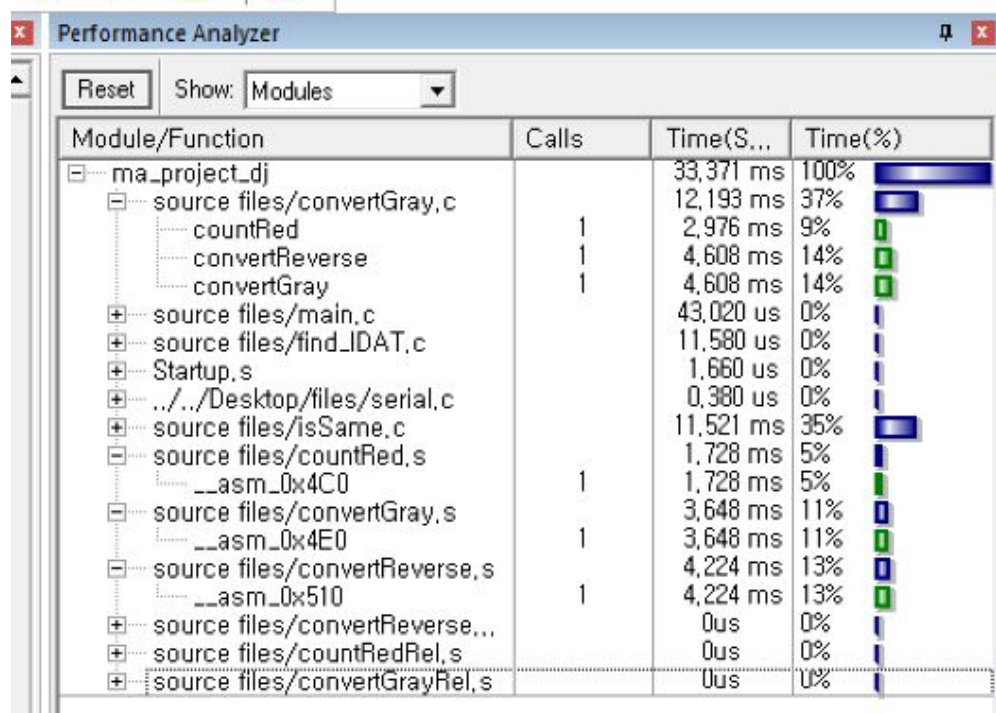
    _sys_exit(0);
}
```

ASSIGN BreakDisable BreakEnable BreakKill BreakList

Disassembly Performance Analyzer

Real-Time Agent: Target Stopped Simulation t1: 0.03337370 sec L39 C1 CAP: NUM SCRL OVR: RAW

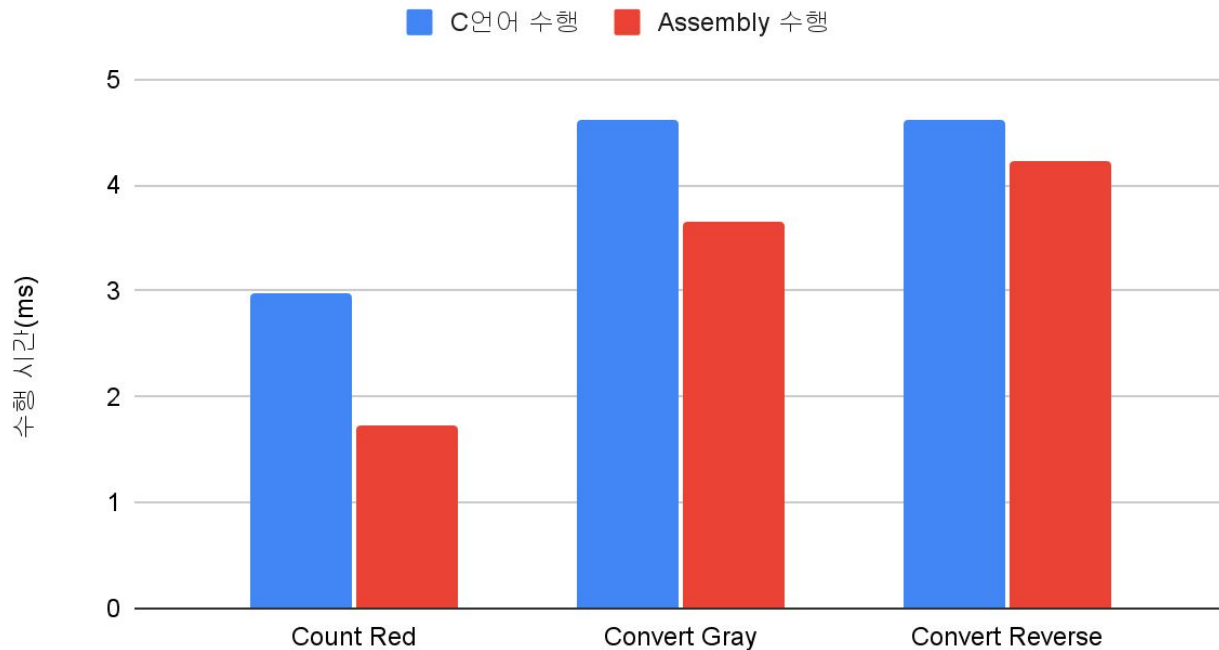
수행시간 비교



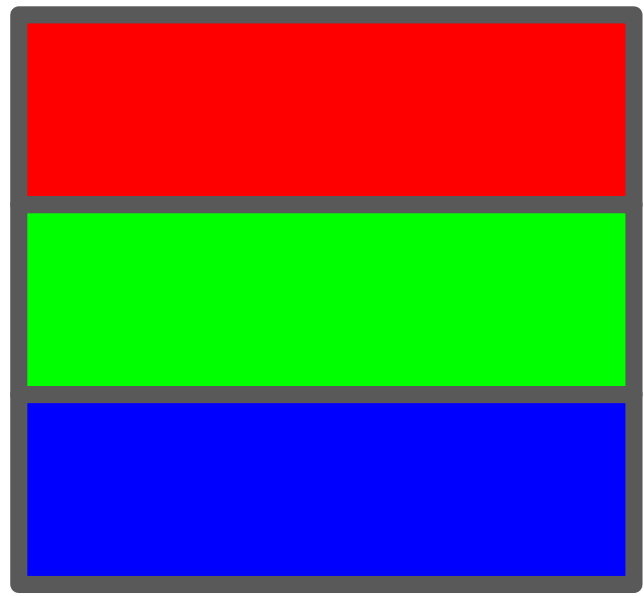
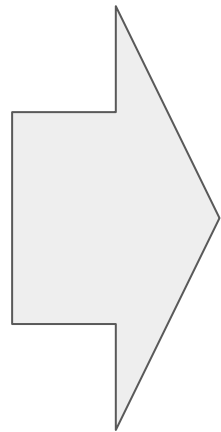
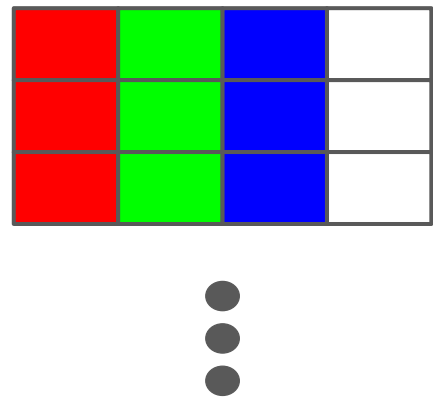
수행시간 비교 시각화

basic		
	C언어 수행 결과	Assembly 최적화 결과
Count Red	2.976	1.728
Convert Gray	4.608	3.648
Convert Reverse	4.608	4.224

C언어 수행 결과, Assembly 최적화 결과



메모리 정렬



reallocation 구현 결과 검증

C:\Users\ehdwn\Desktop\keil_practice\ma_project_dj\ma_project_dj\proj - uVision

File Edit View Project Flash Debug Peripherals Tools SVCS Window Help

Registers

Register	Value
R0	0x00000000
R1	0x00000000
R2	0x00000000
R3	0x00000000
R4	0x30000000
R5	0x30000000
R6	0x30000000
R7	0x30000000
R8	0x30000000
R9	0x30000000
R10	0x30000000
R11	0x40000036
R12	0x00000000
R13 (SP)	0x20000000
R14 (LR)	0x00000000
R15 (PC)	0x00000000
CPSR	0x60000010
SPSR	0x00000000

User/Sys...

- Fast Interrupt
- Interrupt
- Supervisor
- Abort

Memory 1

Address: 0x40000000

0x40000000: 89 50 4E 47 0D 0A 1A 0A
0x40000008: 00 00 00 00 49 48 44 52
0x40000010: 00 00 00 00 00 00 02 80
0x40000018: 08 06 00 00 00 5F 11
0x40000020: 43 00 00 00 01 73 52 47
0x40000028: 42 00 AE CE 1C E9 00 00
0x40000030: 20 00 49 44 41 54 78 5E
0x40000038: 5C BD 77 73 23 59 92 8C
0x40000040: 1B 29 21 08 B2 44 77 8F
0x40000048: 58 71 67 EF B5 77 FF 7B
0x40000050: 0F EF E2 2A 0C 95 8D 0D
error 129: Memory - map size truncated to 128K

Memory 2

Address: 0x30000000

0x30000000: 78 77 92 21 77 67 FF F3
0x30000008: DB 82 67 80 A9 32 C2 F9
0x30000010: B5 33 D1 E8 59 CD 4D CD
0x30000018: 6C F0 74 2C AD AE B4 D8
0x30000020: 88 B6 B4 C6 AD 31 DA 44
0x30000028: 46 EF CF 3E 4D 06 26 79
0x30000030: D8 F0 B1 31 0D B3 CC BF
0x30000038: A3 E7 2B 86 CD 05 D8 B8
0x30000040: B1 DC 4C 6D B1 CD 45
0x30000048: 6C 7B BC B1 13 D6 9D A2
0x30000050: B3 D3 6D 03 5A 35 FA 5C

Memory 3

Address: 0x30004B00

0x30004B00: 5C 23 1B B2 58 B5 DF B3
0x30004B08: 6D 40 F7 1C 92 9E C3 F1
0x30004B10: 17 D2 6D 86 D0 D1 C5 C5
0x30004B18: B6 6D C6 C7 C7 B6 B8 B4
0x30004B20: 14 34 47 6C C4 B8 EF 5D
0x30004B28: FE 4D 6D E5 28 6D 66 97
0x30004B30: F5 03 D6 53 D0 CB DF 79
0x30004B38: 9A 7F 6B F7 FD 31 E3 5A
0x30004B40: 44 5B 6C 37 4B 5B 6B F6
0x30004B48: 3E E7 D8 AC 30 D1 7F 7A
0x30004B50: 13 31 F1 B8 D6 FB 8B FF

Memory 4

Address: 0x30002580

0x30002580: 5E 73 EC 08 0F EF 7B 3C
0x30002588: 9D 02 3F B3 2A 4F C3 F0
0x30002590: 7D C5 B6 35 DD 4F 34 BA
0x30002598: B1 B5 31 D1 32 B1 CD 46
0x300025A0: CB 31 BC D4 4B F4 55 17
0x300025A8: 34 17 0B 7A 13 77 BA E5
0x300025B0: A2 FB C6 4D 9F F0 BA 89
0x300025B8: 69 59 3F 46 18 6C 4D 2E
0x300025C0: 3F CC 46 37 11 2F 1D D3
0x300025C8: 8D E5 C4 2C C3 4E 6F E3
0x300025D0: 9F 1A CF 6B 1F 9A 26 8C

Performance Analyzer

Module/Function	Calls	Time(S...)	Time(%)
ma_project_dj		33.008 ms	100%
source files\lmd_IDAT.c		53.023 us	0%
Startup.s		1.860 us	0%
./././Desktop/files/serial.c		0.380 us	0%
source files\IsSame.c		11.521 ms	35%
source files/main_reallo.c		2.740 us	0%
source files/convertReverse.s		0us	0%
source files/convertGray.s		0us	0%
source files/countRed.s		0us	0%
source files/reAllocation.c		17.570 ms	53%
countRedRel	1	2.784 ms	8%
convertReverseRel	1	6.336 ms	19%
convertGrayRel	1	4.225 ms	13%
reAllocation	1	4.032 ms	12%
source files/convertReverse...		558.620 us	2%
asm_jb544	1	558.620 us	2%
source files/countRedRel.s		660.660 us	2%
asm_jb548	1	660.660 us	2%
source files/convertGrayRel.s		2.833 ms	9%
asm_jb770	1	2.833 ms	9%

main.c main_reallo.c reAllocation.c convertReverse.s convertReverseRel.s convertGrayRel.s countRedRel.s

```
5 #include "convertGray.h"
6 #include "reAllocation.h"
7 #include "isSame.h"
8
9 extern void _sys_exit(int return_code);
10 extern void convertReverseRelasm(uint8_t* origin, uint8_t* toSave, int size);
11 extern int countRedRelasm(uint8_t* origin, int size);
12 extern void convertGrayRelasm(uint16_t* toSave, uint8_t* targetR, uint8_t* targetG, uint8_t* targetB, int
13
14 int main(void) {
15     const int PIXEL_COUNT = 9600;
16
17     uint8_t* pixel_start = find_IDAT( (uint8_t*)0x40000000 ) + 4;
18
19     uint8_t* r = (uint8_t*)0x30000000;
20     uint8_t* g = r + PIXEL_COUNT; // 0x30002580
21     uint8_t* b = g + PIXEL_COUNT; // 0x30004B00
22
23     int* count = (int*)0x30010000;
24
25     uint16_t* grayTarget = (uint16_t*)0x30020000; // ~0x30024B00
26     uint16_t* grayTarget2 = (uint16_t*)0x30030000; // ~0x30024B00
27
28     uint8_t* rRev = (uint8_t*)0x30040000;
29     uint8_t* gRev = rRev + PIXEL_COUNT; // 0x30042580
30     uint8_t* bRev = gRev + PIXEL_COUNT; // 0x30044B00
31
32     uint8_t* rRev2 = (uint8_t*)0x30050000;
33     uint8_t* gRev2 = rRev2 + PIXEL_COUNT; // 0x30052580
34     uint8_t* bRev2 = gRev2 + PIXEL_COUNT; // 0x30054B00
35
36     uint8_t* isSameReturn = (uint8_t*)(count + 2);
37
38     // reallocate data
39     reAllocation(pixel_start, r, g, b, PIXEL_COUNT);
40
41     convertGrayRel(grayTarget, r, g, b, PIXEL_COUNT);
42     convertGrayRelasm(grayTarget2, r, g, b, PIXEL_COUNT);
43     *isSameReturn = isSame_uint16(grayTarget, grayTarget2, PIXEL_COUNT);
44
45     convertReverseRel(r, rRev, PIXEL_COUNT*3);
46     convertReverseRelasm(r, rRev2, PIXEL_COUNT*3);
47     *(isSameReturn+1) = isSame_uint8(rRev, rRev2, PIXEL_COUNT*3);
48
49     *count = countRedRel(r, PIXEL_COUNT);
50     *(count + 1) = countRedRelasm(r, PIXEL_COUNT);
51
52     _sys_exit(0);
53 }
54
55
56
57
```

Real-Time Agent: Target Stopped Simulation t1: 0.03301110 sec L53 C1 CAP_NUM SCRL OVRI R/W

C언어 & ASM 구현 결과 상호 검증(cnt, gray)

The screenshot displays the Keil uVision IDE interface for a project named 'ma_project_dj'. The main window shows the C source code for 'main.c', which includes headers for 'convertGray.h', 'realloc.h', and 'isSame.h'. The code defines a function 'extern void _sys_exit(int return_code);' and a 'main' function that performs a grayscale conversion on a 9600-pixel image. The 'main' function uses a loop to process each pixel, converting it from RGB to grayscale and then back to RGB using a series of arithmetic operations. The code also includes a 'realloc' function to dynamically allocate memory for the pixel data.

The Performance Analyzer window shows the execution time for various modules and functions. The 'ma_project_dj' module takes 33.008 ms to execute. The 'convertGrayRel' function is the most time-consuming, taking 17.370 ms. The 'realloc' function takes 4.032 ms. The 'convertReverseRel' function takes 1.660 ms. The 'countRedRel' function takes 0.830 ms. The 'convertGrayRel' function is called 11,521 times, and the 'realloc' function is called 2,740 times.

The Memory 1 window shows the memory address 0x40000000, which is the start of the pixel data. The Memory 2 window shows the memory address 0x30010000, which is the start of the pixel data. The Memory 3 window shows the memory address 0x30020000, which is the start of the pixel data. The Memory 4 window shows the memory address 0x30030000, which is the start of the pixel data.

The bottom status bar shows the Real-Time Agent is stopped, and the simulation is running. The time is 0:03:01:11.0 sec, and the CPU usage is 155 C1. The CAP MAN, SCRL, and OVR are also shown.

C언어 & ASM 구현 결과 상호 검증 (reverse)

C:\Users\ehdwml\Desktop\Keil_practice\ma_project_dj\ma_project_dj\proj - uVision

File Edit View Project Flash Debug Peripherals Tools SVCS Window Help

Registers

Register	Value
Current	
R0	0x00012C3
R1	0x0000000
R2	0x0000000
R3	0x0000000
R4	0x3000000
R5	0x3001000
R6	0x3004000
R7	0x300C580
R8	0x3003000
R9	0x3005000
R10	0x3001000
R11	0x4000036
R12	0x0000000
R13 (SP)	0x0000000
R14 (LR)	0x0000000
R15 (PC)	0x0000000
CPSR	0x6000000
SPSR	0x0000000

Performance Analyzer

Module/Function	Calls	Time(S...)	Time(%)
ma_project_dj		33.008 ms	100%
source files\find_IDAT.c		53.020 us	0%
Startup.s		1.660 us	0%
./././Desktop/files/serial.c		0.380 us	0%
source files\isSame.c		11.521 ms	35%
source files\main_reallo.c		2.740 us	0%
source files\convertReverse.s		0us	0%
source files\convertGray.s		0us	0%
source files\countRel.s		0us	0%
source files\reAllocation.c		17.370 ms	53%
countRelRel		2.784 ms	8%
convertReverseRel		6.336 ms	19%
convertGrayRel		4.225 ms	13%
reAllocation		4.032 ms	12%
source files\convertReverse...		558.620 us	2%
asm_0x544		558.620 us	2%
source files\countRelRel.s		660.660 us	2%
asm_0x548		660.660 us	2%
source files\convertGrayRel.s		2.833 ms	9%
asm_0x770		2.833 ms	9%

Memory 1

Address	Value
0x40000000	89 50 4E 47 0D 0A 1A 0A
0x40000008	00 00 00 00 49 48 44 52
0x40000010	00 00 00 00 00 00 02 80
0x40000018	08 06 00 00 00 53 F9 11
0x40000020	43 00 00 00 01 73 52 47
0x40000028	42 00 AE CE 1C E9 00 00
0x40000030	20 00 49 44 41 54 78 5E
0x40000038	5C BD 77 73 23 59 92 EC
0x40000040	1B 29 21 08 B2 44 77 8F
0x40000048	58 71 67 EF B5 77 FF 78
0x40000050	8F EF B2 20 85 83 00 00

Memory 2

Address	Value
0x30010000	C3 12 00 00 C3 12 00 00
0x30010008	01 00 00 00 01 00 00 00
0x30010010	00 00 00 00 00 00 00 00
0x30010018	00 00 00 00 00 00 00 00
0x30010020	00 00 00 00 00 00 00 00
0x30010028	00 00 00 00 00 00 00 00
0x30010030	00 00 00 00 00 00 00 00
0x30010038	00 00 00 00 00 00 00 00
0x30010040	00 00 00 00 00 00 00 00
0x30010048	00 00 00 00 00 00 00 00
0x30010050	00 00 00 00 00 00 00 00

Memory 3

Address	Value
0x30040000	87 88 6D DE 88 98 00 0C
0x30040008	22 7D 98 7F 56 CD 3D 06
0x30040010	4A CC 2E 14 76 23 B2 32
0x30040018	93 0F 8B D3 32 51 4B 24
0x30040020	77 49 48 39 52 CE 25 88
0x30040028	89 10 3D C1 B2 F9 89 86
0x30040030	27 0F 4E CE F2 4C 33 A0
0x30040038	5C 18 D4 79 32 FA 21 47
0x30040040	4E 8C 23 B8 92 2E 32 BA
0x30040048	93 84 43 4E EC 29 62 5D
0x30040050	4C 2C 93 FC A5 C3 05 A3

main.c

```
#include "convertGray.h"
#include "reAllocation.h"
#include "isSame.h"

extern void _sys_exit(int return_code);
extern void convertReverseRelasm(uint8_t* origin, uint8_t* toSave, int size);
extern int countRedRelasm(uint8_t* origin, int size);
extern void convertGrayRelasm(uint16_t* toSave, uint8_t* targetR, uint8_t* targetG, uint8_t* targetB, int size);

int main(void) {
    const int PIXEL_COUNT = 9600;

    uint8_t* pixel_start = find_IDAT( (uint8_t*)0x40000000 ) + 4;

    uint8_t* r = (uint8_t*)0x30000000;
    uint8_t* g = r + PIXEL_COUNT; // 0x30002580
    uint8_t* b = g + PIXEL_COUNT; // 0x30004B00

    int* count = (int*)0x30010000;

    uint16_t* grayTarget = (uint16_t*)0x30020000; // ~0x30024B00
    uint16_t* grayTarget2 = (uint16_t*)0x30030000; // ~0x30024B00

    uint8_t* rRev = (uint8_t*)0x30040000;
    uint8_t* gRev = rRev + PIXEL_COUNT; // 0x30042580
    uint8_t* bRev = gRev + PIXEL_COUNT; // 0x30044B00

    uint8_t* rRev2 = (uint8_t*)0x30050000;
    uint8_t* gRev2 = rRev2 + PIXEL_COUNT; // 0x30052580
    uint8_t* bRev2 = gRev2 + PIXEL_COUNT; // 0x30054B00

    uint8_t* isSameReturn = (uint8_t*) (count + 2);

    // reallocate data
    reAllocation(pixel_start, r, g, b, PIXEL_COUNT);

    convertGrayRel(grayTarget, r, g, b, PIXEL_COUNT);
    convertGrayRelasm(grayTarget2, r, g, b, PIXEL_COUNT);
    *isSameReturn = isSame_uint16(grayTarget, grayTarget2, PIXEL_COUNT);

    convertReverseRel(r, rRev, PIXEL_COUNT*3);
    convertReverseRelasm(r, rRev2, PIXEL_COUNT*3);
    *(isSameReturn+4) = isSame_uint8(rRev, rRev2, PIXEL_COUNT*3);

    *count = countRedRel(r, PIXEL_COUNT);
    *(count + 1) = countRedRelasm(r, PIXEL_COUNT);

    _sys_exit(0);
}
```

Real-Time Agent: Target Stopped Simulation t1: 0.03301110 sec L37 C1 CAP: NUM SCRL OVR: RAW

C언어 & ASM 구현 결과 상호 검증 (reverse)

C:\Users\ehdwm\Desktop\Keil_practice\ma_project_dj\ma_project_dj\proj - uVision

File Edit View Project Flash Debug Peripherals Tools SVCS Window Help

Registers

Register	Value
Current	
R0	0x00012C3
R1	0x0000000
R2	0x0000000
R3	0x0000000
R4	0x3000000
R5	0x3001000
R6	0x3004000
R7	0x300C580
R8	0x3003000
R9	0x3005000
R10	0x3001000
R11	0x4000036
R12	0x0000000
R13 (SP)	0x0000000
R14 (LR)	0x0000000
R15 (PC)	0x0000000
CPSR	0x6000000
SPSR	0x0000000

Performance Analyzer

Module/Function	Calls	Time(S...)	Time(%)
ma_project_dj		33.008 ms	100%
source files\find_IDAT.c		53.020 us	0%
Startup.s		1.660 us	0%
./././Desktop/files/serial.c		0.380 us	0%
source files\isSame.c		11.521 ms	35%
source files\main_reallo.c		2.740 us	0%
source files\convertReverse.s		0us	0%
source files\convertGray.s		0us	0%
source files\countRel.s		0us	0%
source files\reAllocation.c		17.370 ms	53%
countRelRel		2.784 ms	8%
convertReverseRel		6.336 ms	19%
convertGrayRel		4.225 ms	13%
reAllocation		4.032 ms	12%
source files\convertReverse...		558.620 us	2%
asm_0x544		558.620 us	2%
source files\countRelRel.s		660.660 us	2%
asm_0x548		660.660 us	2%
source files\convertGrayRel.s		2.833 ms	9%
asm_0x770		2.833 ms	9%

Memory 1

Address: 0x4000000

0x40000000: 89 50 4E 47 0D 0A 1A 0A
0x40000008: 00 00 00 00 49 48 44 52
0x40000010: 00 00 00 00 00 00 02 80
0x40000018: 08 06 00 00 00 53 F9 11
0x40000020: 43 00 00 00 01 73 52 47
0x40000028: 42 00 AE CE 1C E9 00 00
0x40000030: 20 00 49 44 41 54 78 5E
0x40000038: 5C BD 77 73 23 59 92 EC
0x40000040: 1B 29 21 08 B2 44 77 8F
0x40000048: 58 71 67 EF 85 77 FF 78
0x40000050: 8F EF E2 20 85 83 00 00
error 129: Memory - map size truncated on 129000

Memory 2

Address: 0x30010000

0x30010000: C3 12 00 00 C3 12 00 00
0x30010008: 01 00 00 01 00 00 00 00
0x30010010: 00 00 00 00 00 00 00 00
0x30010018: 00 00 00 00 00 00 00 00
0x30010020: 00 00 00 00 00 00 00 00
0x30010028: 00 00 00 00 00 00 00 00
0x30010030: 00 00 00 00 00 00 00 00
0x30010038: 00 00 00 00 00 00 00 00
0x30010040: 00 00 00 00 00 00 00 00
0x30010048: 00 00 00 00 00 00 00 00
0x30010050: 00 00 00 00 00 00 00 00

Memory 3

Address: 0x30040000

0x30040000: 87 88 6D DE 88 98 00 0C
0x30040008: 22 7D 98 7F 56 CD 3D 06
0x30040010: 4A CC 2E 14 76 23 B2 32
0x30040018: 93 0F 8B D3 32 51 48 24
0x30040020: 77 49 48 39 32 CE 25 8B
0x30040028: 89 10 3D C1 B2 F9 89 86
0x30040030: 27 0F 4E CE F2 4C 33 AD
0x30040038: 5C 18 D4 79 32 FA 21 47
0x30040040: 4E 8C 23 B8 92 2E 32 BA
0x30040048: 93 84 43 4E EC 29 62 5D
0x30040050: 4C 2C 93 FC A5 C3 05 A3

Memory 4

Address: 0x30050000

0x30050000: 87 88 6D DE 88 98 00 0C
0x30050008: 22 7D 98 7F 56 CD 3D 06
0x30050010: 4A CC 2E 14 76 23 B2 32
0x30050018: 93 0F 8B D3 32 51 48 24
0x30050020: 77 49 48 39 32 CE 25 8B
0x30050028: 89 10 3D C1 B2 F9 89 86
0x30050030: 27 0F 4E CE F2 4C 33 AD
0x30050038: 5C 18 D4 79 32 FA 21 47
0x30050040: 4E 8C 23 B8 92 2E 32 BA
0x30050048: 93 84 43 4E EC 29 62 5D
0x30050050: 4C 2C 93 FC A5 C3 05 A3

main.c main_reallo.c reAllocation.c convertReverse.s convertReverseRel.s convertGrayRel.s countRedRel.s

```
5 #include "convertGray.h"
6 #include "reAllocation.h"
7 #include "isSame.h"
8
9 extern void _sys_exit(int return_code);
10 extern void convertReverseRelasm(uint8_t* origin, uint8_t* toSave, int size);
11 extern int countRedRelasm(uint8_t* origin, int size);
12 extern void convertGrayRelasm(uint16_t* toSave, uint8_t* targetR, uint8_t* targetG, uint8_t* targetB, int
13
14 int main(void) {
15     const int PIXEL_COUNT = 9600;
16
17     uint8_t* pixel_start = find_IDAT( (uint8_t*)0x40000000 ) + 4;
18
19     uint8_t* r = (uint8_t*)0x30000000;
20     uint8_t* g = r + PIXEL_COUNT; // 0x30002580
21     uint8_t* b = g + PIXEL_COUNT; // 0x30004B00
22
23     int* count = (int*)0x30010000;
24
25     uint16_t* grayTarget = (uint16_t*)0x30020000; // ~0x30024B00
26     uint16_t* grayTarget2 = (uint16_t*)0x30030000; // ~0x30024B00
27
28     uint8_t* rRev = (uint8_t*)0x30040000;
29     uint8_t* gRev = rRev + PIXEL_COUNT; // 0x30042580
30     uint8_t* bRev = gRev + PIXEL_COUNT; // 0x30044B00
31
32     uint8_t* rRev2 = (uint8_t*)0x30050000;
33     uint8_t* gRev2 = rRev2 + PIXEL_COUNT; // 0x30052580
34     uint8_t* bRev2 = gRev2 + PIXEL_COUNT; // 0x30054B00
35
36     uint8_t* isSameReturn = (uint8_t*) (count + 2);
37
38     // reallocate data
39     reAllocation(pixel_start, r, g, b, PIXEL_COUNT);
40
41     convertGrayRel(grayTarget, r, g, b, PIXEL_COUNT);
42     convertGrayRelasm(grayTarget2, r, g, b, PIXEL_COUNT);
43     *isSameReturn = isSame_uint16(grayTarget, grayTarget2, PIXEL_COUNT);
44
45     convertReverseRel(r, rRev, PIXEL_COUNT*3);
46     convertReverseRelasm(r, rRev2, PIXEL_COUNT*3);
47     *(isSameReturn+4) = isSame_uint8(rRev, rRev2, PIXEL_COUNT*3);
48
49     *count = countRedRel(r, PIXEL_COUNT);
50     *(count + 1) = countRedRelasm(r, PIXEL_COUNT);
51
52     _sys_exit(0);
53 }
54
55
56
57
```

Real-Time Agent: Target Stopped Simulation t1: 0.03301110 sec L37 C1 CAP: NUM SCRL OVR: RAW

구조 변화 X 초기 수행 속도

Performance Analyzer

Reset Show: Modules

Module/Function	Calls	Time(S...	Time(%)
ma_project.dj		40,477 ms	100%
source files/find_IDAT.c		53,020 us	0%
Startup.s		1,660 us	0%
.././Desktop/files/serial.c		0,380 us	0%
first grayrel.s		4,032 ms	10%
__asm_0x8C4	1	4,032 ms	10%
first reverserel.s		5,760 ms	14%
__asm_0x8A8	1	5,760 ms	14%
firstcountrel.s		1,728 ms	4%
__asm_0x888	1	1,728 ms	4%
source files/isSame.c		11,521 ms	28%
source files/main_reallo.c		2,740 us	0%
source files/convertReverse.s		0us	0%
source files/convertGray.s		0us	0%
source files/countRed.s		0us	0%
source files/reAllocation.c		17,378 ms	43%
countRedRel	1	2,784 ms	7%
convertReverseRel	1	6,336 ms	16%
convertGrayRel	1	4,225 ms	10%
reAllocation	1	4,032 ms	10%
source files/convertReverse...		0us	0%
source files/countRedRel.s		0us	0%
source files/convertGrayRel.s		0us	0%

최적화 결과 수행시간 비교

Performance Analyzer

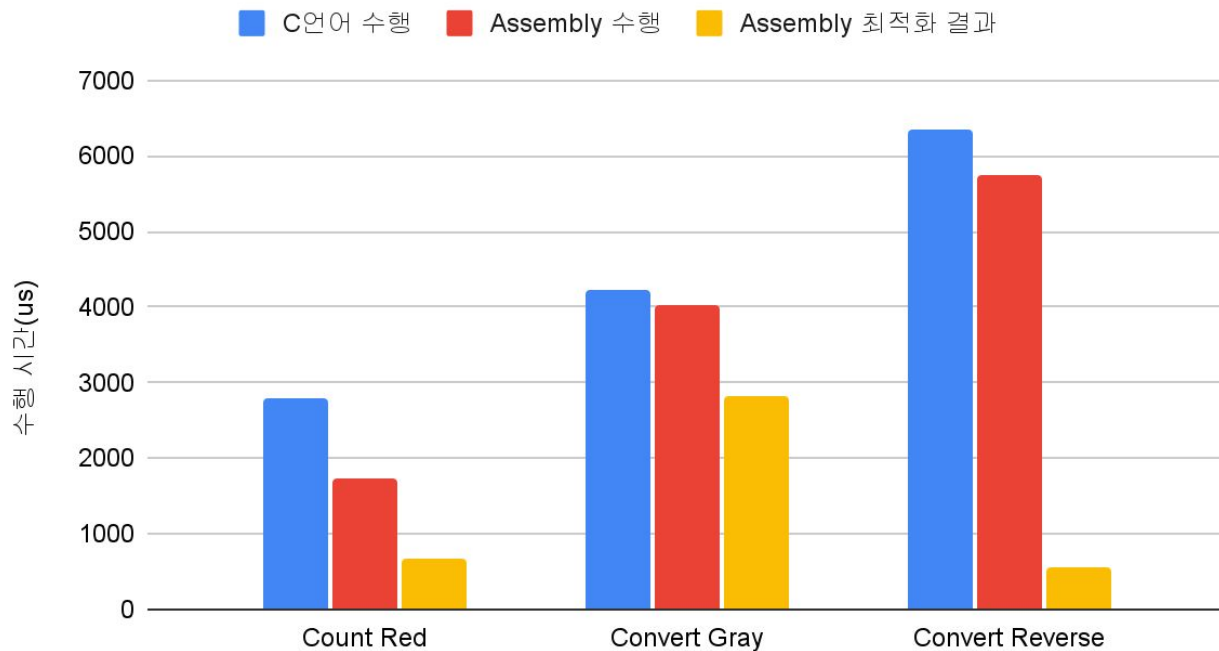
Reset Show: Modules

Module/Function	Calls	Time(S...	Time(%)
ma_project_dj		33,008 ms	100%
source files/find_IDAT.c		53,020 us	0%
Startup.s		1,660 us	0%
.././Desktop/files/serial.c		0,380 us	0%
source files/isSame.c		11,521 ms	35%
source files/main_reallo.c		2,740 us	0%
source files/convertReverse.s		0us	0%
source files/convertGray.s		0us	0%
source files/countRed.s		0us	0%
source files/reAllocation.c		17,378 ms	53%
countRedRel	1	2,784 ms	8%
convertReverseRel	1	6,336 ms	19%
convertGrayRel	1	4,225 ms	13%
reAllocation	1	4,032 ms	12%
source files/convertReverse...		558,620 us	2%
__asm_0x544	1	558,620 us	2%
source files/countRedRel.s		660,660 us	2%
__asm_0x5A8	1	660,660 us	2%
source files/convertGrayRel.s		2,833 ms	9%
__asm_0x770	1	2,833 ms	9%

수행시간 비교 시각화

relocation			
(us)	C언어 수행	Assembly 수행	Assembly 최적화 결과
Count Red	2784	1728	660.66
Convert Gray	4225	4032	2833
Convert Reverse	6336	5760	558.62

C언어 수행 결과, Assembly, 최적화 결과



가속 논리

- 작은 메모리 접근(LDR, STR)은 data bus 점유로 다음 instruction을 느리게 만듦
 - Word, Half word, Byte단위 Load, Store는 모두 같은 사이클 점유
 - LDM, STM을 이용하면 Cycle 절약 가능.
 - 여러 Byte단위(이하 Block 단위)로 Load, Store하여 한 번에 많은 양을 block 단위로 처리
 - Block 단위에 맞지 않는 size의 경우는 일반적인 Byte단위로 처리
- 많은 Loop 횟수는 작은 Branch로 인해 cycle 손해
 - 고정된 Loop는 횟수 반복문이 아닌 정적인 명령으로 설계(Loop Unrolling)
 - Block 단위 설계를 통한 Loop 횟수 절감 (Loop 횟수 = $(\text{Size} / \text{Block}) + (\text{Size} \% \text{Block})$)
- Binary 성질 이용
 - $255 - X = \sim X$
 - if ($X \geq 128$), $X[7] = 1$
 - 위의 성질을 이용하여 4 Byte를 하나의 Register에서 연산.

Rverse 연산 설명

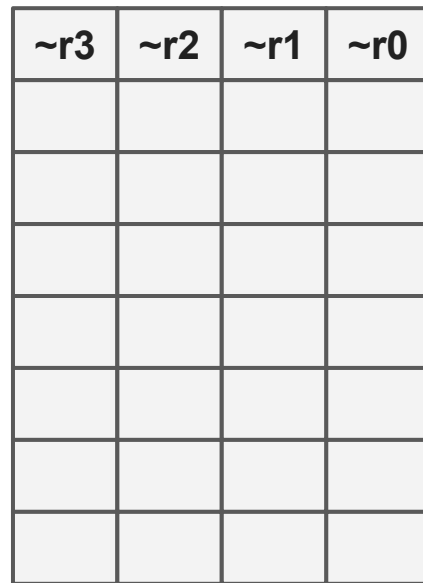
32 Byte 단위 Load,

8번의 word단위 reverse 연산,

32 Byte 단위 Store,

```
if Size % 32 != 0,
```

일반적인 Byte 단위 연산



Convert Gray 설명

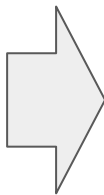
4 word 단위 R, G, B Load

word에서 R, G, B를 1 Byte씩 가져와 2 Byte Gray pixel 생성,

4개의 r, g, b, 정보를 종합, 2Byte의 Gray pixel 정보를 생성,

4개 픽셀(2 word)를 STM을 이용하여 store

3	2	1	0
3	2	1	0
3	2	1	0



1	0	Low
3	2	High

Convert Gray 설명

8 Byte(2 word) 단위 R pixel 정보 Load

Bit mask 사용, 1Byte 단위로 MSB 정보 확인 (TST r0, #128)

Rotate를 이용해서 4pixel 모두 확인

두 word에서 동일한 수행

3	2	1	0
1000_0000	1000_0110	000_1100	1100_0100
0001_0000	0100_0110	000_1100	1110_0100

&

0000_0000	0000_0000	000_0000	1000_0000
-----------	-----------	----------	-----------

전제 performance

Performance Analyzer

Reset Show: Modules

Module/Function	C..	Time(Sec)	Time(%)
ma_project_dj		54,803 ms	100%
source files/reAllocation.c		17,378 ms	32%
countRedRel	1	2,784 ms	5%
convertReverseRel	1	6,336 ms	12%
convertGrayRel	1	4,225 ms	8%
reAllocation	1	4,032 ms	7%
source files/find_IDAT.c		53,020 us	0%
Startup.s		1,660 us	0%
source files/mainnew.c		4,040 us	0%
.././Desktop/files/serial.c		0,380 us	0%
first grayrel.s		0us	0%
first reverserel.s		0us	0%
firstcountrel.s		0us	0%
source files/isSame.c		11,521 ms	21%
source files/convertReverse.s		4,224 ms	8%
__asm_0x510	1	4,224 ms	8%
source files/convertGray.s		3,648 ms	7%
__asm_0x4E0	1	3,648 ms	7%
source files/countRed.s		1,728 ms	3%
__asm_0x4C0	1	1,728 ms	3%
source files/convertGray.c		12,193 ms	22%
countRed	1	2,976 ms	5%
convertReverse	1	4,608 ms	8%
convertGray	1	4,608 ms	8%
source files/convertHeverseRel.s		558,620 us	1%
__asm_0x544	1	558,620 us	1%
source files/countRedRel.s		660,660 us	1%
__asm_0x5A8	1	660,660 us	1%
source files/convertGrayRel.s		2,833 ms	5%
__asm_0x770	1	2,833 ms	5%

Reset

Show:

Modules



Module/Function	C..	Time(Sec)	Time(%)	
[-] ma_project_dj		62,271 ms	100%	
[-] source files/reAllocation.c		17,378 ms	28%	
[-] countRedRel	1	2,784 ms	4%	
[-] convertReverseRel	1	6,336 ms	10%	
[-] convertGrayRel	1	4,225 ms	7%	
[-] reAllocation	1	4,032 ms	6%	
[-] source files/convertGray.c		12,193 ms	20%	
[-] countRed	1	2,976 ms	5%	
[-] convertReverse	1	4,608 ms	7%	
[-] convertGray	1	4,608 ms	7%	
[+] source files/isSame.c		11,521 ms	19%	
[+] first reverserel.s		5,760 ms	9%	
[+] source files/convertReverse.s		4,224 ms	7%	
[+] first grayrel.s		4,032 ms	6%	
[+] source files/convertGray.s		3,648 ms	6%	
[+] source files/countRed.s		1,728 ms	3%	
[+] firstcountrel.s		1,728 ms	3%	
[+] source files/find_IDAT.c		53,020 us	0%	
[+] source files/mainnew.c		4,040 us	0%	
[+] Startup.s		1,660 us	0%	
[+] .././Desktop/files/serial.c		0,380 us	0%	
[+] source files/convertGrayRel.s		0us	0%	
[+] source files/convertReverseRel.s		0us	0%	
[+] source files/countRedRel.s		0us	0%	